Day-Type Structure and Short-Horizon Forecasting of Daily Demand

A preregistered, reproducible data-science study of a 30-day demand series (June 2026)

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Abstract

Background. Operational planning needs to know how much demand rises on marketing-event days and how well tomorrow's demand can be predicted. **Data and design.** We analysed a single daily series of 30 consecutive days (2026-06-01 to 2026-06-30) with marketing-event and holiday flags. The research question, hypotheses, exclusions, tests, effect sizes and robustness checks were preregistered and committed to version control (tag *prereg-v1*) before outcome analysis, and the pipeline was first validated on synthetic fixtures. **Findings.** Mean demand on event days (156.6) exceeded regular days (133.9) by 22.7 units (Welch 95% CI 12.4 to 32.9; Hedges g = 2.24; $p < 0.001$); the result was robust to all four preregistered checks. Because every event day is a Friday or Saturday, this is an association bundled with day-of-week, not a causal marketing lift. A calendar-week effect (upward trend) was also strong. Unsupervised analysis (PCA, K-Means, DBSCAN, Ward) found 3 principal components explaining 86% of feature variance and 4 well-separated clusters (silhouette 0.45) aligned with the weekly rhythm. In rolling-origin one-step-ahead evaluation, a seasonal ARIMA model (RMSE 1.24) and a calendar-feature ridge regression (RMSE 1.75) far outperformed a seasonal-naive baseline (RMSE 6.11). **Implications.** The weekly cycle and drift are highly predictable in this series, which supports weekly staffing and inventory planning; testing marketing on non-weekend days is needed to identify a causal effect. **Limits:** n = 30 and a single, unusually regular series.

Keywords: *preregistration, reproducibility, hypothesis testing, ANOVA, PCA, clustering, SARIMA, time-series forecasting, demand planning.*

1. Introduction and literature background

Demand planning has two recurring questions: *what do promotions and calendar events change?* and *how well can near-term demand be forecast?* Both are easy to answer in ways that look convincing and are wrong when the analysis is flexible enough to fit the data it is testing. Preregistration, which fixes the question, variables and analysis before outcomes are inspected, is a response to this problem (Nosek et al., 2018). Reporting effect sizes with interval estimates rather than significance alone (Cumming, 2014; Wasserstein & Lazar, 2016) addresses a second.

Inference. Group comparisons use Welch's unequal-variance t-test (Welch, 1947) with the Mann-Whitney U test (Mann & Whitney, 1947) as a rank-based counterpart, the Hodges-Lehmann shift as its location estimate (Hodges & Lehmann, 1963), and Hedges' g as a small-sample-corrected standardised effect (Hedges, 1981). Normality is assessed with Shapiro-Wilk (Shapiro & Wilk, 1965) and Kolmogorov-Smirnov tests, the latter with the Lilliefors correction when parameters are estimated (Lilliefors, 1967). Multi-group comparisons use ANOVA with Tukey's HSD (Tukey, 1949).

Structure discovery. Principal component analysis (Hotelling, 1933) compresses correlated features; clusters are sought with K-Means, density-based DBSCAN (Ester et al., 1996) and Ward hierarchical linkage (Ward, 1963), and cluster quality is judged with the silhouette coefficient (Rousseeuw, 1987).

Forecasting. Seasonal ARIMA models follow Box and Jenkins (1970) and the modern treatment of Hyndman and Athanasopoulos (2021). Decomposition uses classical additive and STL methods (Cleveland et al., 1990); stationarity is assessed with the Augmented Dickey-Fuller test (Dickey & Fuller, 1979); accuracy is reported with MAPE, RMSE and MAE (Hyndman & Koehler, 2006), using rolling-origin evaluation (Tashman, 2000). Ridge regression (Hoerl & Kennard, 1970) provides a regularised calendar-feature competitor.

Contribution. This report applies that toolkit to one small series with a deliberately transparent workflow: plan locked first, synthetic dry run second, real analysis third, with all deviations logged. A small case study cannot generalise, but it can show exactly what such a workflow does and does not license one to conclude.

2. Research question and preregistration

Primary question (falsifiable). In the window 2026-06-01 to 2026-06-30, does mean daily demand on marketing-event days differ from mean daily demand on regular days (no event, no holiday)? H_0: $\mu_{event} = \mu_{regular}$; H_A: $\mu_{event} \neq \mu_{regular}$ (expected direction: positive). $\alpha = 0.05$, two-sided. Holiday days (n = 4) were excluded from the primary contrast. The decision rule fixed in advance: Welch's t-test is primary if Shapiro-Wilk does not reject normality in both groups, otherwise Mann-Whitney U; both are always reported.

Identifiability limit found before analysis. A design-only inspection (dates, missingness, flag counts; no outcome summaries) showed that all 8 event days are Fridays or Saturdays and all 4 holidays are Sundays. There are therefore no non-event Fridays/Saturdays to compare against, and the event contrast cannot be separated from a weekend effect. This was written into the preregistration, which restricts interpretation to an association bundled with day-of-week.

Data-access disclosure. Four raw lines were incidentally printed (four Monday–Thursday demand values were visible) before the plan was written; no summary or test of demand was run. **Order of events** (from git history): preregistration, synthetic pipeline tests and environment lockfile committed and tagged *prereg-v1*; then real-data analysis. Two post-lock deviations are logged (a header timestamp correction and a Ljung-Box burn-in bug fix); neither affected any model-selection or test decision.

3. Data and methodology

Data. 30 daily observations (SHA-256 recorded in the preregistration), no missing values, no duplicates, consecutive dates. Variables: *demand* (outcome), *marketing_event* and *holiday* (binary). Derived variables: day type (Event n = 8, Holiday n = 4, Regular n = 18), day of week, calendar week (weeks 1–4 complete Monday–Sunday; days 29–30 form a partial week 5) and a time index.

Component	Specification (all preregistered)
Normality / assumptions	Shapiro-Wilk, KS (+Lilliefors), Brown-Forsythe
Primary test	Welch t and Mann-Whitney U; Hedges g, Hodges-Lehmann (10,000 bootstrap resamples, seed 20260601), permutation test
ANOVA	One-way (Event/Holiday/Regular) + Tukey HSD; two-way week x day type (weeks 1–4, Type II SS)
Robustness	Permutation; OLS + linear trend (HC3); leave-one-week-out; log scale
PCA / clustering	11 engineered features, z-scored; PCs to ≥ 80% variance; K-Means k = 2–6 (silhouette, elbow); DBSCAN; Ward
Time series	Decomposition (period 7); ADF; SARIMA grid by AICc on train (23 days); holdout 7 days; 30-day forecast
Capstone models	Rolling origin (16 one-step forecasts): naive, seasonal naive, SARIMA, Ridge (calendar + lag-1)

Table 1. Analysis plan summary.

Reproducibility. The repository contains raw/interim/processed data folders, executed notebooks, a source package, a pinned environment lockfile (78 packages), an output-contract document and a pytest suite of 18 data-blind tests on synthetic fixtures (planted effect recovery, nominal false-positive rate on null data, contract checks, determinism, and a guard that the pipeline never reads the raw file). All stochastic steps use seed 20260601.

4. Results I: hypothesis tests and ANOVA

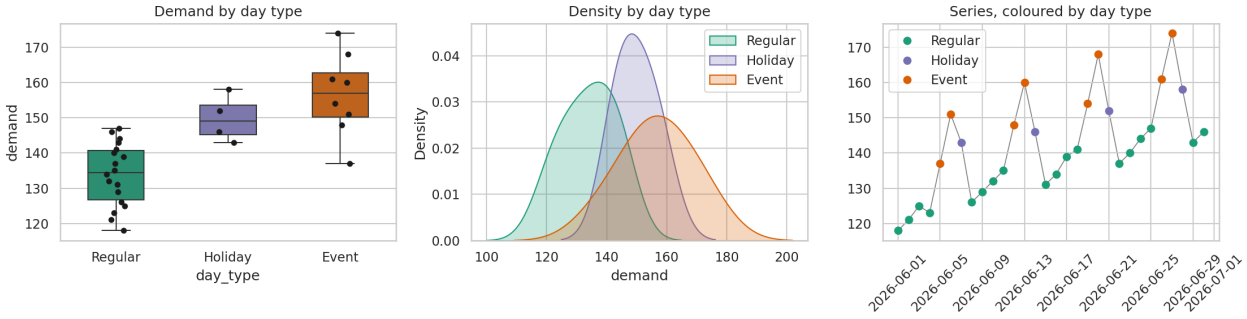


Figure 1. Demand by day type (box and strip plot, density, and time series coloured by day type).

Normality. No group departed detectably from normality (Shapiro-Wilk p = 0.99, 0.80, 0.64 for Event, Holiday, Regular; Lilliefors p ≥ 0.78). Brown-Forsythe did not reject variance homogeneity (p = 0.45). With n = 4 and n = 8 these tests have little power, which is why the nonparametric and permutation analyses are reported alongside.

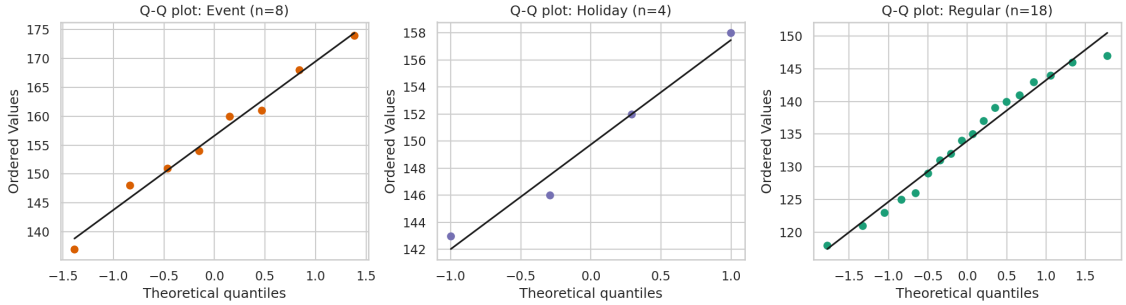


Figure 2. Normal Q-Q plots by day type; points close to the line are consistent with normality (small groups, so low power).

Quantity	Event (n=8)	Regular (n=18)	Estimate [95% CI]
Mean (SD)	156.6 (11.7)	133.9 (8.9)	diff = 22.7 [12.4, 32.9]
Welch t-test	t(10.8) = 4.89	p < 0.001	primary (rule satisfied)
Mann-Whitney U	U = 136.5	p < 0.001	rank-biserial r = 0.90
Hodges-Lehmann shift			22.5 [14.0, 31.5]
Hedges g			2.24 [1.21, 3.27]

Table 2. Primary contrast, Event vs Regular. H_0 is rejected at α = 0.05.

Robustness. The permutation test (p < 0.001), the trend-adjusted OLS estimate (21.6 units, 95% CI 16.1 to 27.0), the log-scale Welch test (p < 0.001; about 16.9% higher) and every leave-one-week-out estimate (21.4 to 24.0) agree in sign and significance. The preregistered criterion for calling the result *robust* is met.

ANOVA. Day type explained a large share of variance: F(2, 27) = 17.3, p < 0.001, η^2 = 0.56 (ω^2 = 0.52); Kruskal-Wallis agrees (p < 0.001). Tukey HSD: Event − Regular = 22.7 (CI 12.7 to 32.7, adjusted p < 0.001); Holiday − Regular = 15.8 (CI 2.8 to 28.8, adjusted p = 0.015); Event − Holiday = 6.9 (CI -7.5 to 21.3, adjusted p = 0.474, not significant). The two-way ANOVA (weeks 1–4, n = 28) shows strong effects of day type (partial η^2 = 0.86) and of calendar week (partial η^2 = 0.75; weekly means 131.1, 139.4, 145.6, 151.6), and no detectable interaction (p = 0.983); the Holiday cell has one observation per week, so the interaction test is weak.

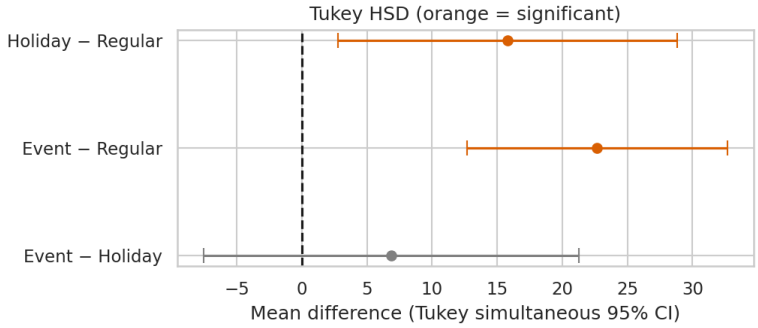


Figure 3. Tukey HSD simultaneous 95% confidence intervals for all day-type pairs (orange = significant).

Reading the result honestly. Day-of-week means run from 131.0 (Mon) to 163.2 (Sat), with Friday (150.0) and Sunday (149.8) in between. Event days are exactly the Friday/Saturday days, so the 22.7-unit contrast is what one expects from the weekly cycle alone as well as from marketing; the data cannot say how much of it is marketing.

5. Results II: dimensionality reduction and clustering

Eleven features (level, two lags, first difference, 3-day rolling mean and SD, sine/cosine of weekday, the two flags, time index) were standardised for the 28 days with complete lag windows. 3 principal components reach 86.1% cumulative variance (PC1 45.4%, PC2 28.2%, PC3 12.5%); the Kaiser rule would retain 4. PC1 is a level-and-trend axis (lags, rolling mean), PC2 an event-day-jump axis (event flag, demand, first difference) and PC3 a holiday/time axis.

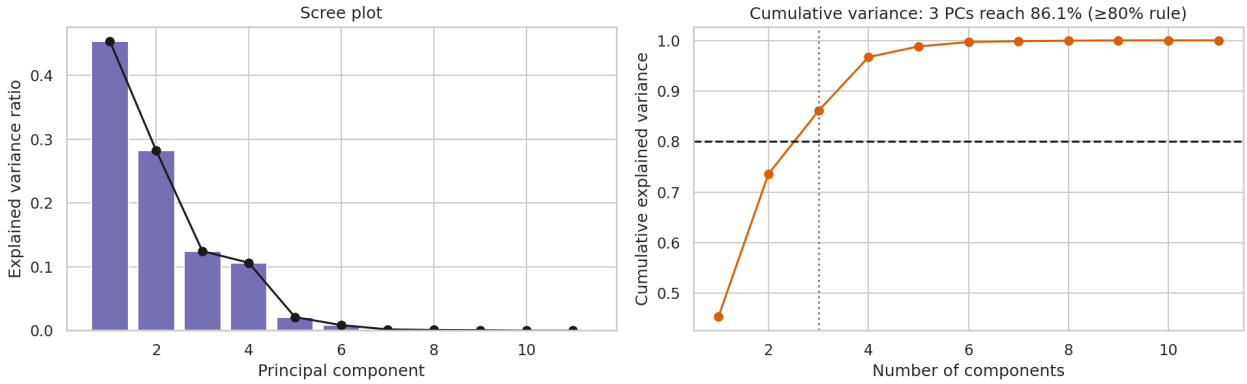


Figure 4. Scree plot and cumulative explained variance; dashed line = 80% retention rule.

K-Means. The mean silhouette is 0.31 (k=2), 0.43 (k=3), 0.45 (k=4), 0.41 (k=5), 0.40 (k=6); the maximum at k = 4 exceeds the preregistered 0.25 threshold, so structure is called meaningful. The inertia curve declines gradually without a sharp elbow, so silhouette is the deciding criterion. DBSCAN (eps from the 3-nearest-neighbour knee) finds four clusters and two noise points; Ward linkage at k = 4 has silhouette 0.44 and cophenetic correlation 0.70. Agreement between methods is high (adjusted Rand index K-Means vs Ward = 0.88, vs DBSCAN = 0.71).

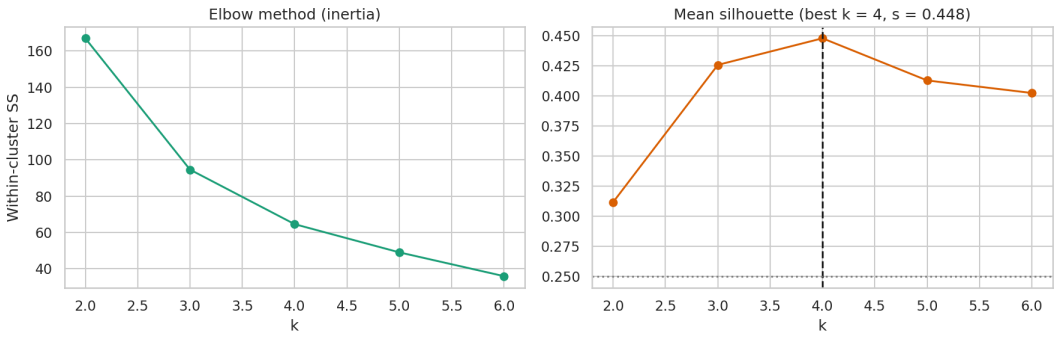


Figure 5. Elbow (inertia) and mean silhouette for K-Means, k = 2–6; dashed line marks the selected k.

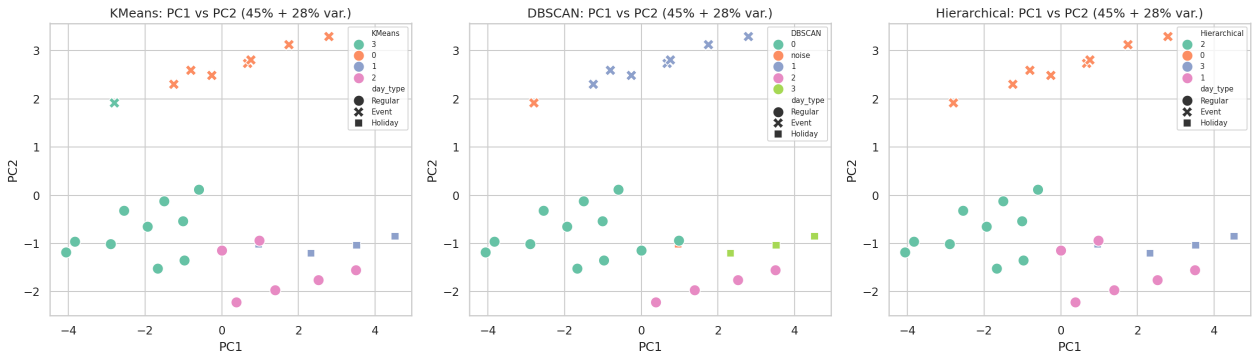


Figure 6. PC1–PC2 projections coloured by K-Means, DBSCAN and Ward clusters (marker shape = day type). An interactive 3D Plotly version is in the repository.

Clusters follow the weekly rhythm: one cluster is almost purely event days (7 of 8), one is exactly the four holidays, and Regular days form two groups (ARI with day type = 0.55). Since flags and weekday phase are among the features this is partly by construction. With 28 rows and 11 correlated features, cluster statistics are unstable and are presented as description, not inference.

6. Results III: time-series analysis and 30-day forecast

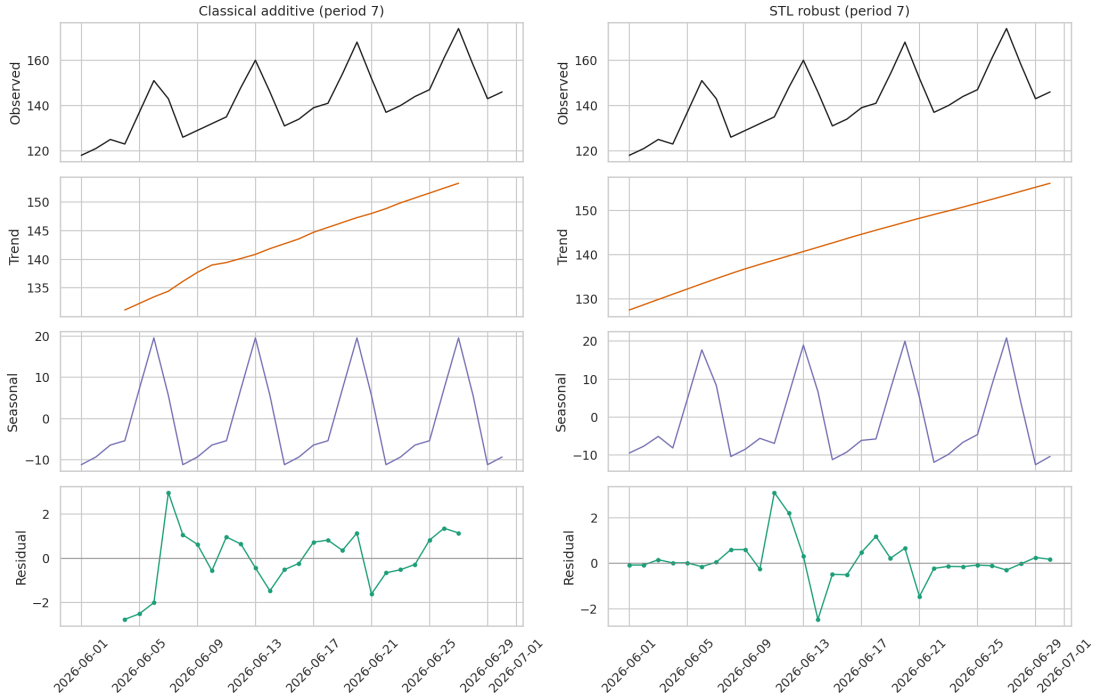


Figure 7. Classical additive and robust STL decomposition (period 7): observed, trend, seasonal, residual.

The series has a pronounced weekly cycle (seasonal range about −11 on Mondays to +20 on Saturdays) on a steadily rising trend (about +6 to +8 units per week), with residuals within about ±3. The ADF test on the level gives p = 0.015 (lag 3), so the preregistered rule sets d = 0; ADF has low power at n = 30 and a constant-only specification, so seasonal differencing (D) was chosen by AICc instead and D = 1 was selected.

Model selection and accuracy. Across 71 converged candidates (training days 1–23), AICc selected SARIMA(1, 0, 0)x(0, 1, 0, 7) (AICc = 79.7; the next model is 2.7 AICc units worse). On the untouched 7-day test split:

Model	MAPE (%)	RMSE	MAE
SARIMA	0.88	1.44	1.35
SeasonalNaive	3.92	6.02	6.00
Naive	8.25	17.03	13.29

Table 3. Holdout accuracy (2026-06-24 to 06-30). The preregistered rule (SARIMA must beat seasonal naive on RMSE) is met.

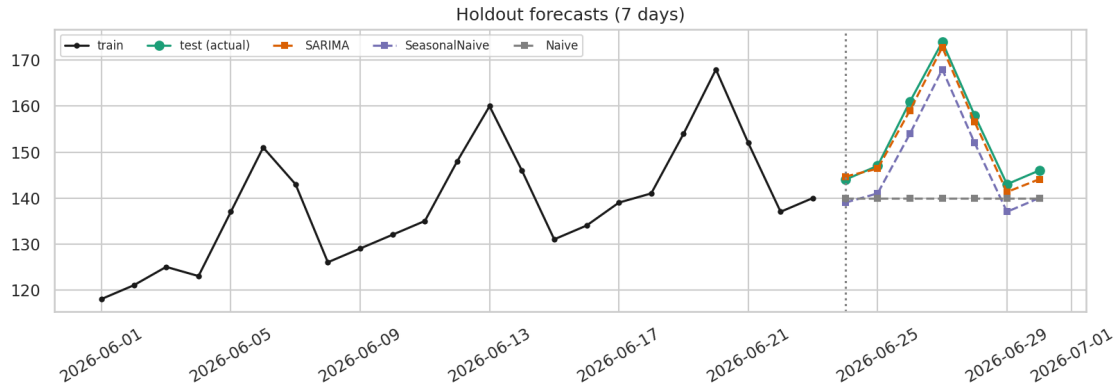


Figure 8. Seven-day holdout: actuals versus SARIMA, seasonal-naive and naive forecasts.

Residual diagnostics are clean (Ljung-Box lag 7, p = 0.58, after discarding differencing burn-in residuals). The seasonal-naive baseline loses mainly because it repeats last week's value and so ignores the weekly upward drift.

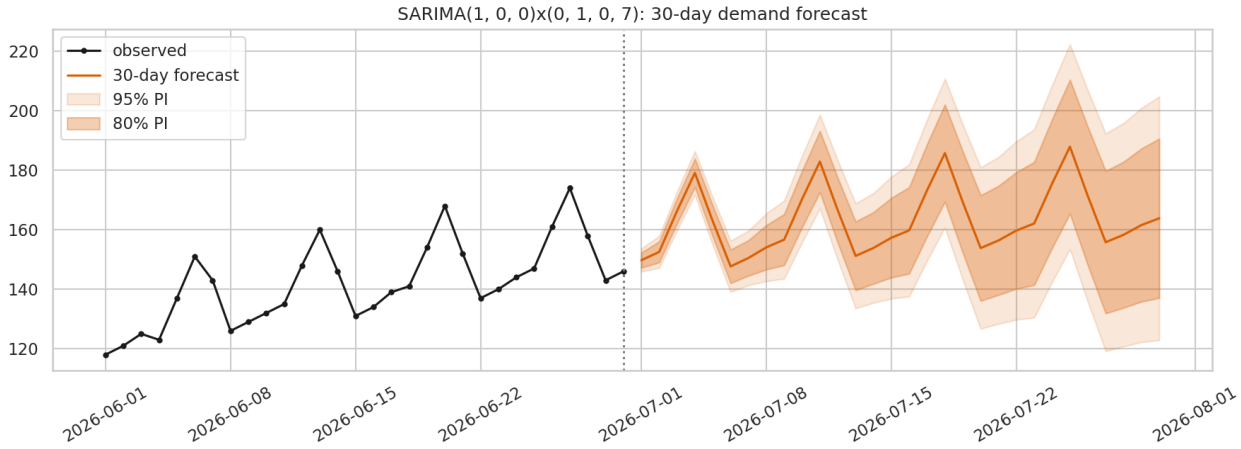


Figure 9. 30-day forecast (2026-07-01 to 07-30) from the model refit on all 30 days, with 80% and 95% prediction intervals.

Forecast. Means of successive 7-day blocks of the forecast range from 158 to 167 units, with Saturdays highest (e.g. 179 on 4 July). The 95% interval half-width grows from 3.9 on day 1 to 41.0 on day 30; beyond about two weeks the interval is wide enough that the point forecast should be treated as a planning scenario, not a commitment.

7. Results IV: predictive modelling workflow (capstone)

Question. Can calendar information and recent history predict next-day demand better than a seasonal-naive baseline? **Design.** Expanding-window one-step-ahead forecasts from 16 origins (target days 15–30), with all tuning (SARIMA specification by AICc; Ridge penalty by time-series cross-validation) repeated inside each training window so that no future information is used.

Model	MAE	RMSE	MAPE (%)	ΔMAE vs seasonal naive [95% CI]
SARIMA	0.97	1.24	0.65	-5.10 [-5.44, -4.67]
Ridge	1.49	1.75	1.01	-4.58 [-5.11, -4.00]
SeasonalNaive	6.06	6.11	4.10	baseline
Naive	9.62	11.17	6.42	3.56 [0.81, 6.19]

Table 4. Rolling-origin results (16 one-step-ahead forecasts). Negative differences = better than baseline (paired bootstrap, 10,000 resamples).

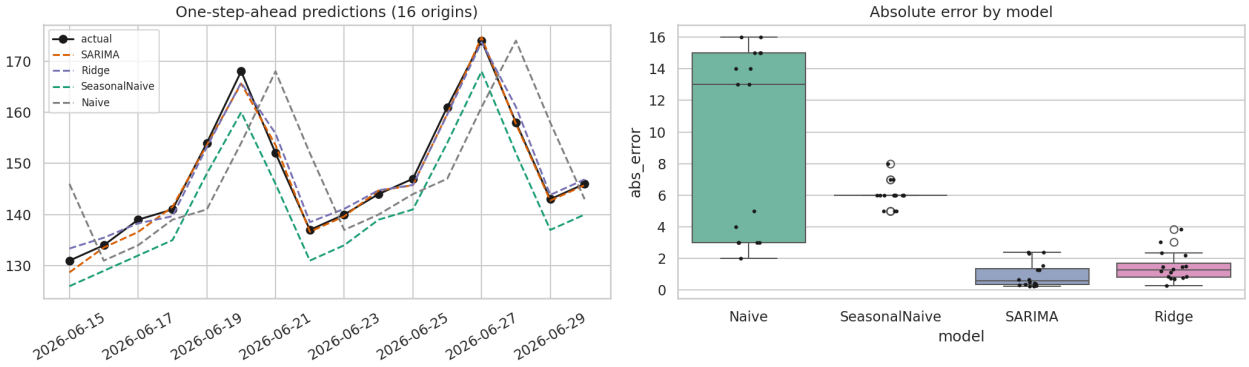


Figure 10. One-step-ahead predictions and absolute error distributions by model.

By the preregistered rule (lowest RMSE) the recommended model is **SARIMA** (RMSE 1.24, MAPE 0.65%), with the calendar-feature Ridge model close behind (RMSE 1.75) and far better than seasonal naive (6.11) and naive (11.17). Ridge is a transparent alternative: it needs no time-series software, and its coefficients show the weekday pattern directly. The near-perfect accuracy of both models says as much about the series as about the models: it is very regular. Performance on longer, noisier data is untested.

8. Business implications

- **Plan capacity by weekday, not by month average.** Saturday demand (163) is about 25% above Monday (131); Friday and Sunday sit in between. A flat weekly staffing or stock plan would under-serve weekends and over-serve early week.
- **Do not credit marketing with the full event-day uplift.** Event days are 22.7 units above regular days, but they are always Fri/Sat. To measure marketing itself, run events on randomly chosen weekdays (including some Mondays–Thursdays and some non-event Fridays/Saturdays) for several weeks; the current calendar cannot identify the effect.
- **Use the forecast as a rolling plan.** One-step-ahead errors are small (about 1–2 units), so day-ahead replenishment can rely on the model; the 30-day horizon should be re-forecast weekly and used with the 80% interval for buffer sizing.
- **Monitor drift.** The series rose by about 6–8 units per week through June. If that growth stops, a model that extrapolates it will over-forecast; track forecast error weekly and retrain if the error exceeds, say, three times its in-sample level.
- **Keep a simple fallback.** Seasonal naive with a weekly drift term is easy to run and audit if the modelling stack is unavailable.

9. Limitations

- **Sample size.** 30 observations from one series; Holiday n = 4; a 7-point test set and 16 strongly autocorrelated forecast origins. All intervals are conditional on the model and are likely optimistic.
- **Confounding by design.** Event ≡ Fri/Sat and holiday ≡ Sunday in this window; no causal claim about marketing is possible. The upward trend further confounds week-to-week comparisons.

- **Unusually regular data.** Residual noise of ±3 on a 130–175 scale suggests clean or synthetic data; performance on noisy real-world demand will be lower.
- **Approximate model selection.** AICc across D = 0 and D = 1 is only approximately comparable at this sample size; ADF has low power; unconstrained SARIMA fits were found to be numerically degenerate and were replaced by constrained fits with a sanity filter (fixed before real-data analysis).
- **Exploratory clustering.** Features include the flags, so agreement with day type is partly built in.
- **Generalisation.** Results describe June 2026 for this series only.

10. Reproducibility and deviations

Everything runs from the repository: `make test` (synthetic pipeline check), `make notebooks` (executes notebooks 01–05 on the real data and writes figures and tables), `make report` (builds this PDF). The preregistration's deviation log records two post-lock entries: (i) a header lock-time typo (05:45 stated, 05:43 actual; the git tag is authoritative) and (ii) a Ljung-Box implementation fix after the first real-data run returned an artefactual $p = 8.6 \times 10^{-6}$ from un-discarded burn-in residuals (corrected $p = 0.58$). One further change was made *before* locking, during the synthetic dry run: constrained SARIMA fitting and the sanity filter, after degenerate intervals appeared on synthetic data. That change was written into the locked plan, so it is not a deviation, but it is disclosed here.

Appendix: supporting tables

Day type	n	Mean	SD	Median	Min	Max
Event	8	156.6	11.7	157.0	137	174
Holiday	4	149.8	6.7	149.0	143	158
Regular	18	133.9	8.9	134.5	118	147

Table A1. Descriptive statistics by day type.

Weekday	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Mean demand	131.0	134.0	135.0	136.5	150.0	163.2	149.8

Table A2. Mean demand by day of week (n = 4 or 5 each).

Component	Eigenvalue	Explained	Cumulative
PC1	5.175	45.4%	45.4%
PC2	3.221	28.2%	73.6%
PC3	1.421	12.5%	86.1%
PC4	1.213	10.6%	96.7%
PC5	0.241	2.1%	98.8%
PC6	0.098	0.9%	99.7%

Table A3. First six principal components of the standardised feature matrix.

SARIMA order	Seasonal order	AICc	AIC	BIC
(1, 0, 0)	(0, 1, 0, 7)	79.69	78.76	80.31
(1, 0, 1)	(0, 1, 0, 7)	82.34	80.34	82.66
(2, 0, 0)	(0, 1, 0, 7)	82.47	80.47	82.79
(1, 0, 0)	(1, 1, 0, 7)	82.74	80.74	83.06
(1, 0, 0)	(0, 1, 1, 7)	82.74	80.74	83.06
(2, 0, 1)	(0, 1, 0, 7)	83.95	80.31	83.40

Table A4. Six best of 71 SARIMA candidates (training window, AICc).

ADF statistic	p-value	Lags	n	5% critical	Differences used
-3.287	0.0155	3	26	-2.981	0 (rule: stationary at 5%)

Table A5. Augmented Dickey-Fuller test on the level series (constant, AIC lag selection, max lag 3).

Date	Forecast	80% PI	95% PI
Wed 01 Jul	149.8	147.2 to 152.3	145.9 to 153.6
Sat 04 Jul	179.1	174.3 to 183.9	171.8 to 186.4
Sat 11 Jul	182.9	172.6 to 193.2	167.2 to 198.6
Sat 18 Jul	185.8	169.4 to 202.1	160.8 to 210.8
Sat 25 Jul	187.9	165.4 to 210.4	153.5 to 222.3
Thu 30 Jul	163.8	137.1 to 190.6	122.9 to 204.8

Table A6. Selected 30-day forecast values (full series in reports/tables/04_forecast_30d.csv).

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